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Expertise-based Adaptive AR Instructions with Motion Data

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Figure 1: We propose a proof-of-concept augmented reality assembly instructions application that adapts the level of detail based on the detected expertise level of the user. As the expertise level decreases, more support is provided by enabling more detailed visualizations: from (a) text and a static 3D model, (b) adding animations, and finally (c) video content as additional information.

Abstract

Augmented Reality has been used to support users in a range of scenarios, often showing advantages over traditional approaches such as paper-based instructions. However, existing systems rarely consider the user's expertise when presenting instructional content. We investigate how expertise can be predicted from data available in a consumer-grade head-mounted display and use these predictions to implement an adaptive augmented reality assembly tutorial that tailors instructional support to the user. Expertise classification is performed on a public dataset using different machine learning algorithms. Results show motion-based expertise classification is feasible, demonstrating the potential for adaptive instructional systems.

Index Terms: Augmented Reality, Adaptive Instructions, Expertise Classification, Machine Learning.

1 Introduction

Advances in portable hardware and head-mounted displays (HMDs) have made augmented reality (AR) widely accessible for both research and consumer applications. One important application area is instructions for assembly [18, 17, 7, 20, 14, 21], maintenance [6], and everyday tasks [23].

A key challenge for AR guidance systems is providing the right amount of assistance without distracting the user. Instructions can range from simple text and images to animated 3D models and avatars. Adaptive tutorials address this challenge by adjusting the level of detail based on factors such as task performance or cognitive load [9, 13, 22]. Previous work has investigated the detection of cognitive load and expertise using data modalities including eye tracking [15], head movements [3], electroencephalogram (EEG) and electrodermal activity (EDA) [1].

Despite these advances, most AR tutorials provide a set of predesigned, fixed instructions throughout the task. This ignores differences in user expertise and can lead to unnecessary mental effort. The expertise reversal effect described by Kalyuga [11] suggests that novice users may struggle when insufficient guidance is provided, while expert users may be hindered by excessive instruction. Consequently, the optimal amount of guidance likely depends on the user's expertise level. To address this issue, we investigate whether expertise can be detected at runtime and used to adapt instructional visualizations accordingly.

Related work [23, 9, 22] highlights the potential of adaptive content, reporting improvements in task completion times and error reduction during task execution. Motivated by these findings, we explore the adaptiveness in the context of AR instructions. Building on previous work in expertise classification [2, 12, 4, 24, 25] and inspired by research on cognitive load detection, both of which have shown promising results using eye-tracking [2, 15, 4, 12], head-tracking [3, 12], and hand-tracking data [25], our goal is to improve the efficiency and user experience of AR assembly tutorials by providing guidance that matches the user's needs.

Therefore, we address the following research question: *Can we predict expertise from data provided by a commercial HMD?*

To answer it, we evaluate different machine learning (ML) algorithms for binary expertise classification on a public dataset, with data that is available from a commercial headset. Based on these findings, we propose an adaptive AR tutorial that automatically adjusts its instruction visualizations according to the user's estimated expertise level, and test it in a pilot study with users.

The main proposed contributions are:

- An evaluation of different ML models for expertise classification from motion data for step-by-step procedures.
- New skill labels for a subset of the widely used Assembly101 [19] public dataset.
- A proof-of-concept AR application for expertise-based adaptive AR assembly tutorials.
- A qualitative pilot study of using the AR application.

2 Expertise Classification

The data that is available from consumer-grade headsets is most commonly body tracking data. Therefore, inspired by the input data that uses head pose information, we utilize the EHTask network proposed by Hu et al. [8], which achieved promising results

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for action classification based on eye and head movements. In addition, we implement the architecture introduced by Fawaz et al. [10], which was originally designed for skill classification on motion data for medical procedures. To provide a broader comparison, we evaluate both deep networks alongside traditional ML algorithms, including Support Vector Machines (SVM), Random Forests (RF), Gradient Boosting (GB), and Multi-Layer Perceptrons (MLP). This is motivated by related work on expertise and cognitive load classification, where these approaches have shown promising performance [4, 1, 9, 12].

Neural Network: EHTask EHTask [8], publicly available[1], uses eye and head movement data to classify user behavior across different tasks in a VR environment. Although the classification task differs from our expertise classification scenario, the network is specifically designed for eye- and head-tracking data and is therefore well suited for our application. For our experiments, we adapt the network to match our data modalities and feature dimensions, focusing on the "Head" module since we aim to use head tracking data. In this work, head movement is represented as linear velocity.

Neural Network: CNN In addition to EHTask, we evaluate the CNN architecture proposed by Fawaz et al. [10]. The model was originally developed for surgical skill assessment [5]. Although our application domain differs, the network is designed for expertise classification using motion data and therefore is highly relevant for our application. As the original implementation is not publicly available, we re-implement the architecture in PyTorch and adapt the input layers to our data modalities.

Traditional Machine Learning Algorithms While deep learning approaches are becoming increasingly popular, traditional ML algorithms remain widely used for expertise and cognitive load classification. Gao et al. [4] evaluate SVM, RF, and LightGBM for expertise detection, while Bhat et al. [1] apply SVM, RF, Extreme Gradient Boosting, and MLP to cognitive load classification. Similar approaches are also used in adaptive tutorials [9] and medical skill assessment [12]. These studies demonstrate that traditional ML methods can achieve competitive performance for both expertise and cognitive load prediction. Therefore, we also evaluate SVMs, RF, GB, and MLP models using the Python sklearn library.

3 Dataset Preparation

3.1 Assembly101

Assembly101 is a widely used object assembly dataset [19]. It contains action annotations at different levels of granularity, skill labels, and 3D pose information of users assembling and disassembling 101 different toys by following paper instructions. The dataset provides multi-view video recordings, timestamps, and camera calibration information.

The dataset uses skill labels ranging from 1 (least skilled) to 5 (most skilled), which are assigned based on assembly speed and the number of mistakes made during the task. The label distribution is highly imbalanced, with only 9% of recordings assigned the lowest skill level and 47% assigned the highest skill level. To reduce the impact of this imbalance, we randomly remove recordings from the original fine-grained split provided by the authors[2] to obtain more balanced data. We downsample the recordings from 60 to 25 FPS, since we plan to stream this over a network and classify in real time.

3.2 Relabelling

After initial experiments with the Assembly101 dataset yielding poor results with all tested models, we investigated the data with a particular focus on the labeling procedure and the skill label distribution. The original labels range from 1 to 5, and are assigned to entire recordings that assemble and disassemble objects in the same session. Since our work focuses on assembly tasks, we re-examined a subset of the videos (157 out of 362) and created new labels, separating assembly and disassembly sequences. Manual relabeling was performed by three annotators with expertise in AR and ML.

Sener et al. [19] state that the skill labels were assigned manually based on assembly speed and the number of mistakes made during the task. However, the exact thresholds used for this process are not publicly available. Since this skill labeling is based on the assumption that speed and number of mistakes are the only indications for skill, we decide to look more closely at the user behavior.

Therefore, the relabeling process considered factors such as hesitation, confidence in handling the object, and the number of mistakes. In addition to assigning an overall skill score, we annotated the transition point between disassembly and assembly and evaluated both segments separately. This allows us to account for users whose performance differs between the two phases of the task. During the relabeling process, we also observed that many participants spend the first few seconds clapping or waiting before beginning the task. Therefore, we remove the first five seconds of each recording in all experiments involving both the original and relabeled data.

Each relabeled recording receives six labels. Three labels correspond to binary expertise classification, distinguishing between novice (0) and expert (1) users, while the remaining three use the original 1-to-5 skill scale. For both labeling schemes, separate labels are assigned to the full recording, the assembly segment, and the disassembly segment. The data is publicly available online[3].

After splitting the recordings into assembly and disassembly segments, we report evaluations on the assembly subset, since this is our chosen scenario. The split per user is the fine-grained-annotations split, which separates the data into training, validation, and test sets, with separate users.

Table 1: Results (classification accuracy), using one sample or majority voting within 10, 20, and 60 seconds windows.

Model	Single Sample	MV10	MV20	MV60
EHTASK	53.2%	54.8%	54.3%	54.5%
SVM	53.71%	54.37%	57.32%	49.35%
RF	52.27%	55.16%	54.27%	63.64%
MLP	51.21%	53.57%	57.93%	59.74%
GB	53.81%	53.57%	57.32%	55.84%
CNN	57.21%	59.42%	60.69%	63.03%

4 Results

The input stream of head velocities is split into windows of 5 seconds, to be classified. Since the tracking can be noisy at times, we also apply a voting scheme after the classifications to accumulate predictions over time for robustness. The outputs are accumulated, and then the majority voting is cast as the predicted class.

The results are shown in Table 1, with the majority voting for windows of 10, 20, and 60 seconds. The best-performing hyperparameters chosen through empirical evaluation are for MLP: hidden layer size = 100, alpha = 0.0001, learning rate = 'constant', for GB: n estimators = 100, learning rate = 0.01, max. depth = 5 subsample = 1.0, for SVM: RBF kernel, regularization parameter C = 1.0, for RF: 100 trees, and default Python sklearn library parameters.

The best accuracy results are achieved with the RF model and our reimplementation of the CNN proposed by Fawaz et al. [10]. Since the CNN model achieves the highest performance in all voting windows, we choose this for further experiments. In comparison, results on original data did not surpass 55% accuracy for any model. Predicting expertise from motion data alone is a challenging task. The available motion data provide only an indirect measure of

[1] https://github.com/CraneHzm/EHTask

[2] https://github.com/assembly-101/assembly101-annotations

[3] https://osf.io/kqugx/overview?view_only=b038ddde0fc14178be846d3fb1b159c8

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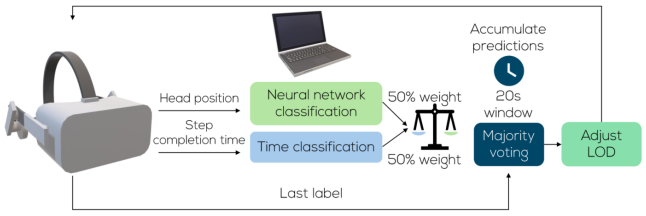


Figure 2: AR application: Data from the headset is streamed to the PC, where NN classification is combined with temporal thresholding and accumulated for 20 seconds, to inform the LOD adjustment.

expertise and may be affected by sensor noise, tracking inaccuracies, and uncertainty in the assigned expertise labels.

5 AR Application

We implement an AR application to assemble a toy from the Assembly101 dataset (excavator) that uses real-time classification and adapts the levels of detail (LODs) of the instructions based on the predicted expertise level. The application offers four different LOD that users can explore based on the selected difficulty setting: very easy, easy, medium, and difficult. These difficulty levels correspond directly to increasing levels of instructional detail.

The highest LOD (4) provides an animated virtual model visualizing the current instruction step, a video demonstrating the current action, and a textual description of the task. LOD 3 removes the video instruction, while LOD 2 replaces the animated virtual model with a static one and also omits the video (see Figure 1). The lowest LOD (1) consists solely of textual instructions.

A similar structure is employed by Huang et al. [9] in their adaptive system. Their approach uses five LODs, including a level that provides no assistance at all. For our use case, however, we considered textual instructions to be the minimum level of support necessary. Furthermore, while their most detailed LOD employs an animated avatar to indicate the required action, we opted to use video instructions instead. This decision was motivated by differences in the application context, namely tasks involving larger machines, whereas our assembly object consists of small individual parts. The remaining visualizations used in our system, such as textual instructions and animated 3D models, are comparable to those presented in their work. The use of animated visualizations is inspired by Radkowski et al. [16], who investigate the effects of different visualization techniques in AR instruction systems in detail.

The application was developed using Unity 2022.3.21f and a Meta Quest 3 headset, which was selected because it is a lightweight HMD that provides access to body tracking data through its API, making it well-suited for AR instruction tasks. To enable communication between the AR application and the classification model, we implemented a simple UDP-based client-server architecture. The Unity application running on the Meta Quest 3 acts as the client collecting head-motion data, while a separate server application executes on a desktop PC (Lenovo Legion 5 with an AMD Ryzen 5 5600H processor and an NVIDIA GeForce RTX 3070 GPU) connected to the same local network. The server receives the recorded user data, performs the classification, and returns the resulting expertise prediction to the AR application.

The system combines two approaches: a simple time-based classifier, which labels users based on whether their average step completion time exceeds an 8-second threshold, and a CNN trained on head-velocity data from Assembly101 dataset. To improve robustness, predictions are aggregated over 20 seconds and combined through a majority voting scheme (see Figure 2). The 8-second threshold was computed as the rounded average step length of the relabeled Assembly101 for assembly of the excavator toy. The resulting expertise label is used to adapt the LOD. Starting in LOD 3, the application gradually increases or decreases the level by 1 during runtime (novice prediction increases the LOD, experts prediction decreases LOD). Difficulty levels are adjusted only when the votes exceed 70%.

6 Pilot Study

6.1 Procedure

Seven participants (4 male, 3 female) aged 24–35 years (M = 27.3, SD = 3.8) were recruited from a local university. All participants reported prior experience with XR, with four indicating slight to moderate familiarity and three reporting high familiarity. Regarding assembly experience, five participants reported some experience with toy assembly, while two reported extensive experience.

Participants first received an introduction to the task, followed by a short familiarization phase to get acquainted with the AR interaction techniques on a few LEGO blocks. They then completed the assembly task while the system automatically classified their expertise and adapted the level of instructional detail for ten assembly steps. They continued the final four assembly steps with manual control over the instruction difficulty, allowing them to explore and compare the different visualization levels at their own request. Upon completing the assembly task, participants took part in a semi-structured interview about their experience with the system.

6.2 Qualitative Results

Visualizations Participants distinguished between the visualization they found most helpful and the one they preferred to use in an AR instruction application. While four of seven participants considered the instructional video the most helpful, only two selected it as their preferred visualization. One participant explained this preference by stating: “The video [was helpful], because you also saw the hands of a user doing the assembly, which helped to understand the process more.” This suggests that video provides an intuitive and natural representation of the assembly process. Three participants preferred the animated 3D model, with several users noting that its usefulness could be improved through greater visual accuracy and detail, particularly for small components such as screws. Most participants (5/7) reported that the visualizations were not distracting, and all participants stated that they relied on all types of visualizations during the assembly task.

LOD Adaptation The adaptive LOD was generally received positively, with five of seven participants reporting that the automatic adaptation was helpful. However, only two participants explicitly noticed the adaptation during the task. Both felt that the transition to the most detailed instruction level occurred at an appropriate moment. One participant commented: “At one point I wasn’t sure on which side the screw was supposed to go and then the video started for the next step and I could see exactly what I was supposed to do.” Only one participant explicitly preferred manual control over the LOD and did not consider the automatic adaptation necessary. This participant primarily relied on the least detailed instruction level and questioned their novice classification, explaining: “Because at the beginning I was just still figuring out the system. And not trying to get on the part fast enough.” This suggests that familiarization with the interface may not have been sufficient, or that the time classification assumption might not accurately reflect the user’s expertise.

Manual LOD selection When given the option to manually change it, some participants preferred having all visualizations available at all times, with one explaining: “Difficulty 1 because none of it was distracting and I prefer to have all of them activated and use whatever I want.” Others preferred adjusting the level of detail individually for each assembly step. One participant suggested

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extending the adaptive system by considering the difficulty of individual assembly steps, providing more guidance for challenging actions and less for simpler ones. Another participant summarized the balance between assistance and simplicity: "I think it was simple enough so that I'm not distracted, but helpful enough if I wasn't confident in what I was doing." Overall, participants responded positively to the system, with comments such as "I really liked the tool as it was." and "It was pretty fun. I really enjoyed this."

7 Conclusion and Future Work

Our experiments suggest that motion data contains information that may be useful for estimating user expertise, although the classification performance remains limited and may be influenced by multiple factors, including inter-user variability and the quality and nature of the expertise labels, as both self-assessments and expert annotations can introduce bias. Future work could investigate additional neural network architectures specifically designed for this task, as well as improved approaches to labeling. Overall, our results provide preliminary evidence that motion-based expertise classification is feasible in the studied scenarios, but further research is needed to improve its reliability and accuracy.

The findings of our pilot study suggest that users were generally not distracted by the automatic adaptation of the LOD in this context, received the automatically adaptive AR tutorial positively, and enjoyed performing the assembly task. In future work, we plan to investigate these findings through a more extensive user study, including an improved user classification approach and a comparison against a standard non-adaptive baseline. We also plan to explore additional datasets, potentially incorporating more modalities available on consumer-grade hardware, and use them for model training.

Finally, future work should also consider the implications of collecting and processing user motion data. While expertise detection can improve the personalization of AR systems, the ability to infer user characteristics from behavioral data raises privacy concerns. Future systems should therefore incorporate privacy-preserving methods and transparent, user-controlled data practices.

Acknowledgments

The work was supported by VRVis GmbH funded in COMET (911654), a program managed by FFG.

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